

DIPAK B PATIL

ML Engineer | AI/ML, Computer Vision & NLP

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SUMMARY

Aspiring ML Engineer with hands-on experience across computer vision, NLP, and Generative AI, including a LangChain/Hugging Face LLM chatbot, a PyTorch-based object detection system, and deployed NLP classifiers. Comfortable owning the full ML lifecycle from data preprocessing through model deployment, with a strong SQL/EDA foundation for data-driven decision-making.

EDUCATION

B.Tech — Computer Science & Engineering (Artificial Intelligence & Machine Learning)

2022 – 2026 | Jalgaon

G.H. Rasoni College of Engineering and Management

TECHNICAL SKILLS

Programming: Python, SQL (MySQL) — joins, subqueries, aggregations, stored procedures

ML/DL: Supervised & unsupervised learning, classification, regression, clustering, model evaluation, feature engineering

Libraries & Frameworks: Pandas, NumPy, Scikit-learn, PyTorch, TensorFlow, LangChain, Matplotlib, Seaborn, Plotly

Generative AI / LLM: Prompt engineering, output parsing, LLM API integration, embeddings, Hugging Face

Data & Reporting: EDA, statistical analysis, Power BI, DAX, dashboarding, data storytelling

Engineering: Git/GitHub, Jupyter Notebook, Google Colab, Streamlit app deployment

PROJECTS

AI-Powered Road Damage Detection using Drone Imagery

Computer Vision, PyTorch, Object Detection

- Improved road defect detection accuracy, measured by mAP, IoU, and F1-score, by training a transfer-learning-based object detection model in PyTorch with data augmentation across varied road conditions.
- Increased detection reliability across lighting and terrain variations, measured through Precision/Recall evaluation, by tuning hyperparameters and validating on annotated drone-imagery datasets.

Text Emotion & Toxicity Detection

NLP, Machine Learning, Deep Learning

- Improved multi-class emotion and toxicity classification performance, measured by F1-score, by fine-tuning LSTM and SVM models on TF-IDF and lemmatized text embeddings.
- Enabled real-time content-moderation decisions, measured by inference response time, by deploying the trained model as an interactive Streamlit application.

AI Chatbot using LangChain, Hugging Face & Streamlit

Generative AI, LLM Integration

- Improved response reliability and structure, measured by consistency of parsed outputs, by implementing prompt templates and structured output parsers on Hugging Face/NVIDIA LLM integrations.
- Delivered an interactive Generative AI interface, measured by end-to-end usability, by integrating LLM APIs and embeddings into a Streamlit front end.

Hotel Booking Cancellation Analysis

Python, EDA, Power BI

- Identified key drivers of booking cancellations across 119,000+ records, measured through univariate, bivariate, and multivariate analysis, by cleaning and analyzing lead time, ADR, and customer-segment data.
- Improved dashboard usability for stakeholders, measured by clarity of cancellation-trend visualizations, by building conditional-column DAX logic in Power BI.

Smart-Wallet: Multi-User Personal Finance Web App

Python, OOP, HTML, CSS

- Enabled secure, isolated financial tracking across multiple users, measured by consistent data segregation between accounts, by architecting an OOP-based multi-user wallet system with authentication.
- Reduced manual expense-tracking effort, measured by automated report generation, by building monthly financial reporting for budgeting and spend analysis.

CERTIFICATES

- Python Programming
- Exploratory Data Analysis
- Power BI
- MySQL